Unconditional monotone size and depth dichotomies for Boolean-valued constraint satisfaction

Bruno P. Cavalar, Igor C. Oliveira

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Allowed standard prerequisites (editorial).

- Known monotone lower bounds, logspace algorithms and threshold circuit bounds. Exact author prior theorem statements retained with citations. Full author depth-reduction/padding support proofs included.

Source reference. Original reference 3: number-theory/sources/ccc-2023-29/native/main.tex, source lines 954–955. Classification overview compares to companion circuit-separation section; no premise from that section is required beyond separately retained exact prior theorem.

Source reference. Original reference 3.2: number-theory/sources/ccc-2023-29/native/main.tex, source lines 1119–1120. Side discussion compares the delivered classification to a separate graph separation, not used in classification proof.

3 Preliminaries

3.1 Notation

Boolean functions. We denote by Mono the set of all monotone Boolean functions. We define $\mathsf{poly} = \left\{n \mapsto n^C : C \in \mathbb{N}\right\}$. A Boolean function $f : \{0,1\}^{\binom{n}{2}} \to \{0,1\}$ is said to be a graph property if $f(G) = f(H)$ for any two isomorphic graphs G and H. Let $\mathcal{F} = \{f_n\}_{n\in\mathbb{N}}$ be a sequence of graph properties, where f_n is defined over undirected graphs on n vertices. We say that $\mathcal{F}$ is *preserved under homomorphisms* if, whenever there is a homomorphism from a graph G to a graph H, we have $\mathcal{F}(G) \leq \mathcal{F}(H)$. We denote by $\mathsf{HomPreserving}$ the set of all graph properties which are preserved under homomorphisms. Note that $\mathsf{HomPreserving} \subseteq \mathsf{Mono}$.

Boolean circuits. We denote by $\mathsf{AC}^0_d[s]$ the family of Boolean functions computed by size-s, depth-d Boolean circuits with unbounded fan-in $\{\wedge, \vee\}$-gates and input literals from $\{x_1, \overline{x_1}, \ldots, x_n, \overline{x_n}\}$. We write $\mathsf{AC}^0[s]$ as a shorthand for $\bigcup_{d=1}^{\infty} \mathsf{AC}^0_d[s]$, and AC^0 as a shorthand of $\mathsf{AC}^0[n^{O(1)}] = \mathsf{AC}^0[\mathsf{poly}]$. We will also refer to $\mathsf{AC}^0_d[\mathsf{poly}]$ by AC^0_d. We write $\mathsf{DNF}[s]$ to denote the family of Boolean functions computed by size-s DNFs, where size is measured by number of terms. We write $\mathsf{CNF}[s]$ analogously. We write $\mathsf{SIZE}[s]$ to denote the family of Boolean functions computed by size-s circuits. We write $\mathsf{DEPTH}[d]$ to denote the family of Boolean functions computed by fan-in 2 circuits of depth d. We denote by $\mathsf{AC}^0[\oplus]$ the family of Boolean functions computed by polynomial-size AC^0 circuits with unbounded fan-in $\oplus$-gates.

We denote by L the family of Boolean functions computed by logspace machines, and by NL the family of Boolean functions computed by polynomial-time nondeterministic logspace machines. Moreover, we denote by $\oplus\mathsf{L}$ the family of Boolean functions computed by polynomial-time nondeterministic logspace machines with a *parity* acceptance condition (i.e., an input is accepted if the number of accepting paths is odd).

Circuit complexity. Given a circuit class $\mathcal{C}$, we write $\mathsf{m}C$ to denote the monotone version of $\mathcal{C}$. Given a function f, we write $\mathsf{mSIZE}(f)$ to denote the size of the smallest monotone circuit computing f and $\mathsf{mDEPTH}(f)$ to denote the smallest depth of a fan-in 2 monotone circuit computing f. Given two Boolean functions f, g, we write $f \leq_m^{\mathsf{mProj}} g$ if there exists a many-one reduction from f to g in which each bit of the reduction is a monotone projection[1] of the input.

Miscellanea. Let $\alpha \in \{0,1\}^n$. We define $|\alpha|_1 := \sum_{i=1}^n \alpha_i$. We call $|\alpha|_1$ the *Hamming weight* of α. We let $\mathrm{supp}(\alpha) = \{i \in [n] : \alpha_i = 1\}$. We let $\mathrm{THR}_{k,n} : \{0,1\}^n \to \{0,1\}$ be the Boolean function such that $\mathrm{THR}_{k,n}(x) = 1 \iff |x|_1 \geq k$.

3.2 Background results

The next lemma, which is proved via a standard "guess-and-verify" approach, shows that nondeterministic logspace computations can be simulated by circuits of size 2^{n^ε} and of depth $d = O_\varepsilon(1)$.

Lemma 3.1 (Folklore; see, e.g., [AHM+08, Lemma 8.1])**.** *For all $\varepsilon > 0$, we have $\mathsf{NL} \subseteq \mathsf{AC}^0[2^{n^\varepsilon}]$.*

Theorem 3.2 ([GKRS19])**.** *There exists $\varepsilon > 0$ such that $\oplus\mathsf{L} \cap \mathsf{Mono} \not\subseteq \mathsf{mSIZE}[2^{o(n^\varepsilon)}]$. Moreover, this separation is witnessed by* 3-XOR-SAT*.*

Theorem 3.3 (Polynomial-size constant-depth vs. larger monotone depth)**.** *For every $k \geq 1$, we have $\mathsf{AC}^0 \cap \mathsf{Mono} \not\subseteq \mathsf{mDEPTH}[(\log n)^k]$. Moreover, this separation holds for a monotone graph property.*

Theorem 3.4 (Polynomial-size constant-depth vs. larger monotone size)**.** *For every $k \geq 1$, we have $\mathsf{AC}^0[\oplus] \cap \mathsf{Mono} \not\subseteq \mathsf{mSIZE}[2^{(\log n)^k}]$.*

Recall that [GKRS19] showed the existence of a monotone function f^{GKRS} in NC^2 that is not in $\mathsf{mSIZE}[2^{n^{\Omega(1)}}]$. As opposed to classical results [Raz85b, And85, AB87, Tar88] that rely on the approximation method, their monotone circuit lower bound employs a lifting technique from communication complexity. It is thus natural to consider if their approach can be adapted to provide a monotone function g that is efficiently computable by constant-depth circuits but is not in $\mathsf{mSIZE}[\mathsf{poly}]$.

As remarked in [GKRS19, dRGR22], all monotone lower bounds obtained from lifting theorems so far also hold for monotone encodings of constraint satisfaction problems (CSPs). Next, we introduce a class of monotone Boolean functions $\mathsf{CSP\text{-}SAT}_S$ which capture the framework and lower bound of [GKRS19].

Encoding CSPs as monotone Boolean functions. Let $R \subseteq \{0,1\}^k$ be a relation. We call k the *arity* of R. Let $V = (i_1, \ldots, i_k) \in [n]^k$, and let $f_{R,V} : \{0,1\}^n \to \{0,1\}$ be the function that accepts a string $x \in \{0,1\}^n$ if $(x_{i_1}, \ldots, x_{i_k}) \in R$. We call $f_{R,V}$ a *constraint application* of R on n variables. (A different choice of the sequence V gives a different constraint application of R.) If S is a finite set of Boolean relations, we call any set of constraint applications of relations from S on a fixed set of variables an *S-formula.* In particular, we can describe an S-formula through a set of pairs (V, R). We say that

[1] A monotone projection is a projection without negations.

an S-formula F is *satisfiable* if there exists an assignment to the variables of F which satisfies all the constraints of F.

Let $S = \{R_1, \ldots, R_k\}$ be a finite set of Boolean relations. Let ℓ_i be the arity of the relation R_i. Note that there are n^{ℓ_i} possible constraint applications of the relation R_i on n variables. Let $N := \sum_{i=1}^{k} n^{\ell_i}$. We can identify each S-formula F on a fixed set of n variables with a corresponding string $w^F \in \{0,1\}^N$, where $w_j^F = 1$ if and only if the j-th possible constraint application (corresponding to one of the N pairs (V, R)) appears in F. Let $\mathsf{CSP\text{-}SAT}_S^n : \{0,1\}^N \to \{0,1\}$ be the Boolean function which accepts a given S-formula F if F is *unsatisfiable*. Note that this is a monotone function. When n is clear from the context or we view $\{\mathsf{CSP\text{-}SAT}_S^n\}_{n \geq 1}$ as a sequence of functions, we simply write $\mathsf{CSP\text{-}SAT}_S$.

Theorem 3.5 (Limits of the direct approach via lifting and CSPs)**.** *Let S be a finite set of Boolean relations. The following holds.*

1. *If $\mathsf{CSP\text{-}SAT}_S \notin \mathsf{mSIZE}[\mathsf{poly}]$ then $\mathsf{CSP\text{-}SAT}_S$ is $\oplus\mathsf{L}$-hard under $\leq_m^{\mathsf{AC}^0}$ reductions.*
2. *If $\mathsf{CSP\text{-}SAT}_S \notin \mathsf{mNC}^1$ then $\mathsf{CSP\text{-}SAT}_S$ is L-hard under $\leq_m^{\mathsf{AC}^0}$ reductions.*

Theorem 3.6. *For every $i \geq 1$, we have $\mathsf{AC}^0 \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \not\subseteq \mathsf{mDEPTH}[(\log n)^i]$. In particular, we have $\mathsf{AC}^0 \cap \mathsf{Mono} \cap \mathsf{GraphProperties} \not\subseteq \mathsf{mNC}^i$.*

Theorem 3.7 ([BGW99])**.** *Any monotone formula computing $\mathsf{OddFactor}_n$ has size $2^{\Omega(n)}$, and any monotone circuit computing $\mathsf{OddFactor}_n$ has size $n^{\Omega(\log n)}$.*

Theorem 3.8 (Dichotomies for the monotone complexity of Boolean-valued CSPs)**.** *Let S be a finite set of Boolean relations. The following holds.*

1. Monotone Size Dichotomy: *If $\mathsf{Pol}(S) \subseteq \mathrm{L}_3$ there is $\varepsilon > 0$ such that $\mathsf{mSIZE}(\mathsf{CSP\text{-}SAT}_S) = 2^{\Omega(n^\varepsilon)}$. Otherwise, $\mathsf{mSIZE}(\mathsf{CSP\text{-}SAT}_S) = n^{O(1)}$.*
2. Monotone Depth Dichotomy: *If $\mathsf{Pol}(S) \subseteq \mathrm{L}_3$ or $\mathsf{Pol}(S) \subseteq \mathrm{V}_2$ or $\mathsf{Pol}(S) \subseteq \mathrm{E}_2$, there is $\varepsilon > 0$ such that $\mathsf{mDEPTH}(\mathsf{CSP\text{-}SAT}_S) = \Omega(n^\varepsilon)$. Otherwise, $\mathsf{CSP\text{-}SAT}_S \in \mathsf{mNC}^2$.*

4 Monotone Complexity of Constraint Satisfaction Problems

In this section, we study the monotone complexity of Boolean-valued CSPs. Our goal is to classify which types of Boolean CSPs are hard for monotone circuit size and monotone circuit depth, eventually proving Theorems 3.5 and 3.8.

We will first spend some time recalling standard definitions and concepts in the theory of CSPs (Section 4.1), as well as a few results about CSPs that were proved in previous works [Sch78, Jea98, BCRV03, BCRV04, ABI+09] (Section 4.2). We will then prove Theorem 3.8 in Section 4.3, and we will finally prove Theorem 3.5 in Section 4.5 after proving some auxiliary results in Section 4.4.

4.1 Definitions

For a good introduction to the concepts defined below, we refer the reader to [BCRV03, BCRV04]. We also refer the reader to Section 3.2 for the definition of the family of functions $\mathsf{CSP\text{-}SAT}_S$, as well as the terms *constraint application, S-formula* and *satisfiable formula*.

We denote by $p_i^n : \{0,1\}^n \to \{0,1\}$ the i-th *projection function* on n variables, whose operation is defined as $p_i^n(x) = x_i$. For a set of Boolean functions B, we denote by $[B]$ the *closure* of B, defined as follows: a Boolean function f is in $[B]$ if and only if $f \in B \cup \{\text{Identity}\}$ or if there exists $g \in B$ and $h_1, \ldots, h_k$ such that $f = g(h_1, \ldots, h_k)$, where each h_i is either a projection function or a function from $[B]$. We can equivalently define $[B]$ as the set of all Boolean functions that can be computed by circuits using the functions of B as gates. Note that $[B]$ necessarily contains an infinite number of Boolean functions, since $p_1^n \in [B]$ for every $n \in \mathbb{N}$; moreover, the constant functions are not necessarily in $[B]$. We say that B is a *clone* if $B = [B]$. A few prominent examples of clones are the set of all Boolean functions (equal to $[\{\wedge, \neg\}]$), monotone functions (equal to $[\{\wedge, \vee, 0, 1\}]$), and linear functions (equal to $[\{\oplus, 1\}]$).

Remark 4.1. *The set of all clones forms a lattice, known as* Post's lattice*, under the operations* $[A] \sqcap [B] := [A] \cap [B]$ *and* $[A] \sqcup [B] := [A \cup B]$*. From the next section onwards, we will refer to the clones defined in [BCRV03] (such as* I_0*,* I_1*, etc.), assuming the reader is familiar with them. For the unfamiliar reader, we refer to Section 7 and Figures 1 and 4, which contain all the definitions of the clones we will need, as well as the entire Post's lattice in graphical representation.*

To avoid confusion, we will always refer to clones with normal-Roman *font (e.g.,* $\mathrm{S}_1, \mathrm{I}_0$*, etc).*

Let S be a finite set of Boolean relations. We denote by $\mathsf{CNF}(S)$ the set of all S-formulas. We denote by $\mathsf{COQ}(S)$ the set of all relations which can be expressed with the following type of formula φ:

$$\varphi(x_1, \ldots, x_k) = \exists y_1, \ldots, y_\ell \, \psi(x_1, \ldots, x_k, y_1, \ldots, y_\ell),$$

where $\psi \in \mathsf{CNF}(S)$. The relations in $\mathsf{COQ}(S)$ will also be referred as *conjunctive queries* over S. We denote by $\langle S \rangle$ the set of relations defined as $\langle S \rangle := \mathsf{COQ}(S \cup \{=\})$. If $S = \langle S \rangle$, we say that S is a *co-clone*. We define

$$\mathsf{CSP} = \{\mathsf{CSP\text{-}SAT}_S : S \text{ is a finite set of relations}\}.$$

We say that $\mathsf{CSP\text{-}SAT}_S$ is *trivial* if $\mathsf{CSP\text{-}SAT}_S$ is a constant function.

Let R be a k-ary Boolean relation and let $f : \{0,1\}^\ell \to \{0,1\}$ be a Boolean function. For $x \in R$ and $i \in [k]$, we denote by $x[i]$ the i-th bit of x.

Definition 4.2. *We say that* f *is a* polymorphism *of* R*, and* R *is* an invariant of f*, if, for all* $x_1, \ldots, x_\ell \in R$*, we have*

$$(f(x_1[1], \ldots, x_\ell[1]), f(x_2[2], \ldots, x_\ell[2]), \ldots, f(x_k[k], \ldots, x_\ell[k])) \in R.$$

We denote the set of all polymorphisms of R *by* $\mathsf{Pol}(R)$*. For a set of relations* S*, we denote by* $\mathsf{Pol}(S)$ *the set of Boolean functions which are polymorphisms of all the relations of* S*. For a set of Boolean functions, we denote by* $\mathsf{Inv}(B)$ *the set of all Boolean relations which are invariant under all functions of* B *(i.e.,* $\mathsf{Inv}(B) = \{R : B \subseteq \mathsf{Pol}(R)\}$*).*

The following summarises the important facts about clones, co-clones and polymorphisms that are relevant to the study of CSPs [JCG97].

Lemma 4.3. *Let* S *and* S' *be sets of Boolean relations and let* B *and* B' *be sets of Boolean functions. We have*

(i) $\mathsf{Pol}(S)$ *is a clone and* $\mathsf{Inv}(B)$ *is a co-clone;*

(ii) If $S \subseteq S'$*, then* $\mathsf{Pol}(S') \subseteq \mathsf{Pol}(S)$*;*

(iii) If $B \subseteq B'$*, then* $\mathsf{Inv}(B') \subseteq \mathsf{Inv}(B)$*;*

(iv) $\mathsf{COQ}(\mathsf{COQ}(S)) = \mathsf{COQ}(S)$*;*

(v) If $S \subseteq S'$*, then* $\mathsf{COQ}(S) \subseteq \mathsf{COQ}(S')$*;*

(vi) $\mathsf{Inv}(\mathsf{Pol}(S)) = \langle S \rangle$*;*

(vii) $\mathsf{Pol}(\mathsf{Inv}(B)) = [B]$*.*

We now define different types of reductions. We say that a reduction is a *monotone OR-reduction* if every bit of the reduction is either constant or can be computed by a monotone disjunction on the input variables. We write $f \leq_m^{\mathsf{mOR}} g$ if there exists a many-one monotone OR-reduction from f to g. We also write $f \leq_m^{\mathsf{AC}^0} g$ if there exists a many-one AC^0 reduction from f to g, and $f \leq_m^{\mathsf{mNL}} g$ if there exists a many-one mNL reduction from f to g[2]. Unless otherwise specified, every reduction we consider will generate an instance of polynomial size on the length of the input.

Finally, we denote by OR^k and NAND^k the k-ary OR and NAND relations, respectively.

[2] A many-one AC^0 (resp. mNL) reduction is one in which each bit of the reduction is either constant or can be computed with a polynomial-size AC^0 circuit (resp. monotone nondeterministic branching program). Recall that a *monotone nondeterministic branching program* is a directed acyclic graph G with two distinguished vertices s and t, in which each edge e is labelled with an input function $\rho_e \in \{1, x_1, \ldots, x_n\}$. Given an input x, the program accepts if there exists a path from s to t in the subgraph G_x of G in which an edge e appears if $\rho_e(x) = 1$.

4.2 Basic facts about CSP-SAT

We state here basic facts about the CSP-SAT function. These facts are proved in the original paper of Schaefer [Sch78], as well as in later papers [Jea98, BCRV03, BCRV04, ABI+09].

Lemma 4.4 below is one of the most important lemmas of this section and will be used many times. It states that $\mathsf{Pol}(S)$ characterises the monotone complexity of $\mathsf{CSP\text{-}SAT}_S$, in the sense that the sets of relations with few polymorphisms give rise to the hardest instances of CSPs. A non-monotone version of this result was proved in [Jea98, BCRV04, Theorem 2.4], and we check in Section 6.2 that their proofs also hold in the monotone case.

Lemma 4.4 (Polymorphisms characterise the complexity of CSPs [Jea98, BCRV04, Theorem 2.4])**.** *If* $\mathsf{Pol}(S_2) \subseteq \mathsf{Pol}(S_1)$*, then* $\mathsf{CSP\text{-}SAT}^n_{S_1} \leq^{\mathsf{mNL}}_m \mathsf{CSP\text{-}SAT}^{\mathsf{poly}(n)}_{S_2}$.

Theorem 4.5 gives monotone circuit upper bounds for some instances of $\mathsf{CSP\text{-}SAT}_S$. Non-monotone variants of this upper bound were originally obtained in the seminal paper of Schaefer [Sch78], and we again check that the monotone variants work in Section 6.3.

Theorem 4.5 (Monotone version of the upper bounds for CSP-SAT [Sch78, ABI+09])**.** *Let S be a finite set of relations. The following holds.*

1. *If* $\mathrm{E}_2 \subseteq \mathsf{Pol}(S)$ *or* $\mathrm{V}_2 \subseteq \mathsf{Pol}(S)$*, then* $\mathsf{CSP\text{-}SAT}_S \in \mathsf{mSIZE}[\mathsf{poly}]$.
2. *If* $\mathrm{D}_2 \subseteq \mathsf{Pol}(S)$*, or* $\mathrm{S}_{00} \subseteq \mathsf{Pol}(S)$*, or* $\mathrm{S}_{10} \subseteq \mathsf{Pol}(S)$*, then* $\mathsf{CSP\text{-}SAT}_S \in \mathsf{mNL}$.

Finally, we state here a result of [ABI+09], which classifies the *non-monotone* complexity of $\mathsf{CSP\text{-}SAT}_S$ under $\leq^{\mathsf{AC}^0}_m$ reductions. The classification of the complexity of $\mathsf{CSP\text{-}SAT}_S$ is based solely on $\mathsf{Pol}(S)$. See Figure 1 for a graphical representation.

Theorem 4.6 (Refined classification of CSP problems [ABI+09, Theorem 3.1])**.** *Let S be a finite set of Boolean relations. The following holds.*

- *If* $\mathrm{I}_0 \subseteq \mathsf{Pol}(S)$ *or* $\mathrm{I}_1 \subseteq \mathsf{Pol}(S)$*, then* $\mathsf{CSP\text{-}SAT}_S$ *is trivial.*
- *If* $\mathsf{Pol}(S) \in \{\mathrm{I}_2, \mathrm{N}_2\}$*, then* $\mathsf{CSP\text{-}SAT}_S$ *is* $\leq^{\mathsf{AC}^0}_m$*-complete for* NP.
- *If* $\mathsf{Pol}(S) \in \{\mathrm{V}_2, \mathrm{E}_2\}$*, then* $\mathsf{CSP\text{-}SAT}_S$ *is* $\leq^{\mathsf{AC}^0}_m$*-complete for* P.
- *If* $\mathsf{Pol}(S) \in \{\mathrm{L}_2, \mathrm{L}_3\}$*, then* $\mathsf{CSP\text{-}SAT}_S$ *is* $\leq^{\mathsf{AC}^0}_m$*-complete for* $\oplus$L.
- *If* $\mathrm{S}_{00} \subseteq \mathsf{Pol}(S) \subseteq {\mathrm{S}_{00}}^2$ *or* $\mathrm{S}_{10} \subseteq \mathsf{Pol}(S) \subseteq {\mathrm{S}_{10}}^2$ *or* $\mathsf{Pol}(S) \in \{\mathrm{D}_2, \mathrm{M}_2\}$*, then* $\mathsf{CSP\text{-}SAT}_S$ *is* $\leq^{\mathsf{AC}^0}_m$*-complete for* NL.
- *If* $\mathsf{Pol}(S) \in \{\mathrm{D}_1, \mathrm{D}\}$*, then* $\mathsf{CSP\text{-}SAT}_S$ *is* $\leq^{\mathsf{AC}^0}_m$*-complete for* L.
- *If* $\mathrm{S}_{02} \subseteq \mathsf{Pol}(S) \subseteq \mathrm{R}_2$ *or* $\mathrm{S}_{12} \subseteq \mathsf{Pol}(S) \subseteq \mathrm{R}_2$*, then either* $\mathsf{CSP\text{-}SAT}_S \in \mathsf{AC}^0$ *or* $\mathsf{CSP\text{-}SAT}_S$ *is* $\leq^{\mathsf{AC}^0}_m$*-complete for* L.

4.3 A monotone dichotomy for CSP-SAT

In this section, we prove Theorem 3.8. We first prove Part (1) of the theorem (the dichotomy for circuit size), and then we prove Part (2) of the theorem (the dichotomy for circuit depth).

Dichotomy for circuits. To prove the dichotomy for circuits, we first show that, for any set of relations S whose set of polymorphisms is contained in L_3, we can monotonically reduce 3-XOR-SAT to $\mathsf{CSP\text{-}SAT}_S$.

Lemma 4.7. *Let S be a finite set of relations. If* $\mathsf{Pol}(S) \subseteq \mathrm{L}_3$*, then* $\text{3-XOR-SAT} \leq^{\mathsf{mNL}}_m \mathsf{CSP\text{-}SAT}_S$.

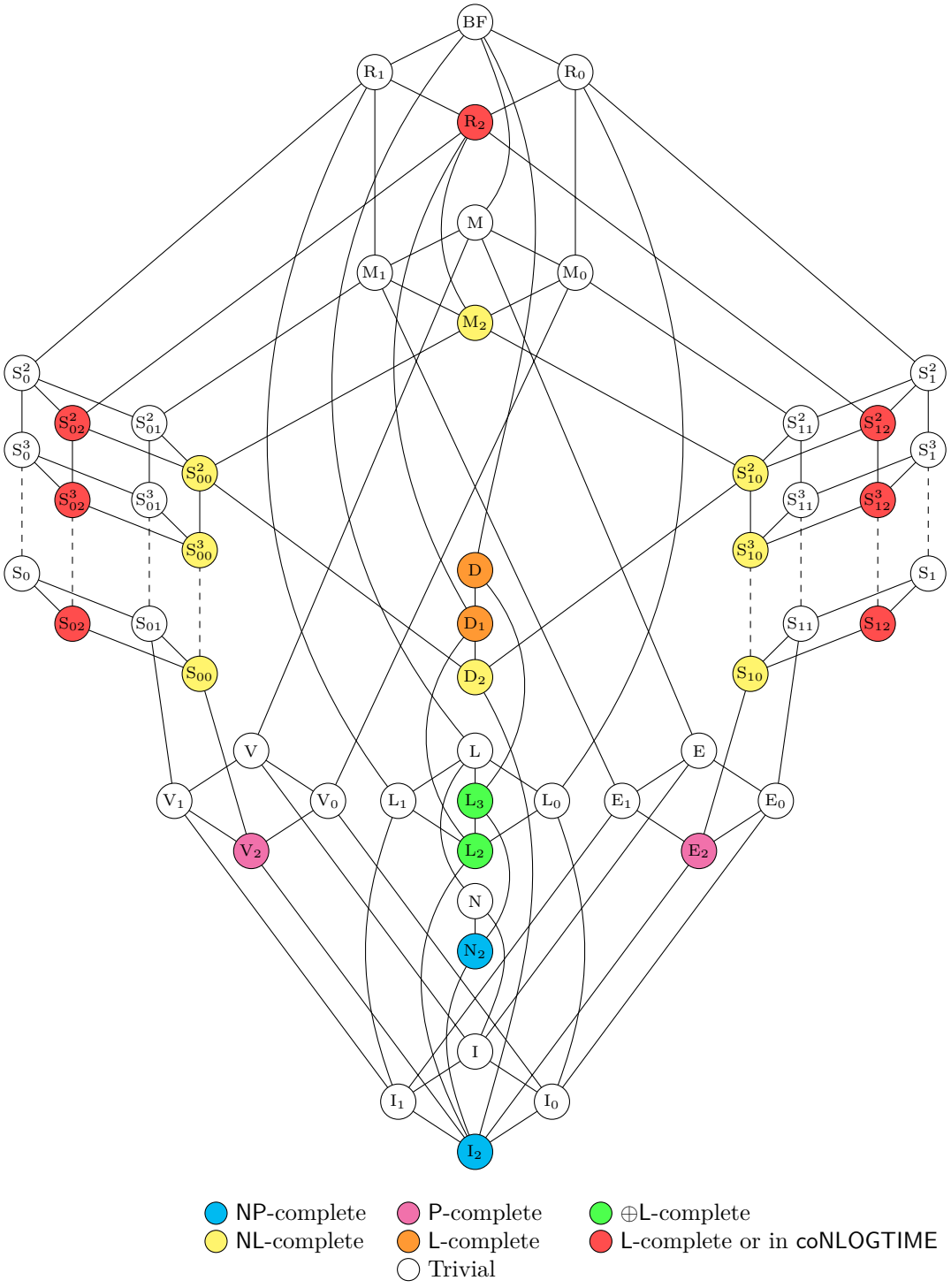


Figure 1: Graph of all closed classes of Boolean functions. The vertices are colored with the complexity of deciding CSPs whose set of polymorphisms corresponds to the label of the vertex. Trivial CSPs are those that correspond to constant functions. Every hardness result is proved under $\leq_m^{\mathsf{AC}^0}$ reductions. See Theorem 4.6 for details. A similar figure appears in [ABI+09, Figure 1].

Proof. Inspecting Post's lattice (Figure 1), note that the only clones strictly contained in L_3 are $\mathrm{L}_2, \mathrm{N}_2$ and I_2. We will first show that the reduction holds for the case $\mathsf{Pol}(S) = \mathrm{L}_2$ and then prove that the reduction also holds for the case $\mathsf{Pol}(S) = \mathrm{L}_3$. Lemma 4.4 will then imply the cases $\mathsf{Pol}(S) \in \{\mathrm{N}_2, \mathrm{I}_2\}$, since $\mathrm{I}_2 \subseteq \mathrm{N}_2 \subseteq \mathrm{L}_3$.

It's not hard to check that, if $\mathsf{Pol}(S) = \mathrm{L}_2$, then $\mathsf{Pol}(S) \subseteq \mathsf{Pol}(\mathsf{3\text{-}XOR\text{-}SAT})$ (it suffices to observe that bitwise XORing three satisfying assignments to a linear equation gives rise to a new satisfying assignment to the same equation). Therefore, from Lemma 4.4 we deduce that $\mathsf{3\text{-}XOR\text{-}SAT}$ admits a reduction to $\mathsf{CSP\text{-}SAT}_S$ in mNL. In order to prove the case $\mathsf{Pol}(S) = \mathrm{L}_3$, we first prove the following claim.

Claim ([ABI+09, Lemma 3.11])**.** *Let S be a finite set of relations such that $\mathsf{Pol}(S) = \mathrm{L}_2$. There exists a finite set of relations S' such that $\mathsf{Pol}(S') = \mathrm{L}_3$ and $\mathsf{CSP\text{-}SAT}^n_S \leq^{\mathsf{mProj}}_m \mathsf{CSP\text{-}SAT}^{n+1}_{S'}$.*

Proof. We describe the proof of Lemma 3.11 in [ABI+09] and observe that it gives a monotone reduction.

For a relation $R \in S$, let $R' = \{(\neg x_1, \ldots, \neg x_k) : (x_1, \ldots, x_k) \in R\}$. Let also $S' = \{R' : R \in S\}$. It's not hard to check that $\mathsf{Pol}(S') = \mathrm{L}_3$, since S' is an invariant of L_2 and N_2, and L_3 is the smallest clone containing both L_2 and N_2; moreover, if $\rho \in \mathsf{Pol}(S')$ and ρ is a Boolean function on at least two bits, then $\rho \in \mathsf{Pol}(S) = \mathrm{L}_2$.

Now let F be a instance of $\mathsf{CSP\text{-}SAT}^n_S$. For every constraint $C = R(x_1, \ldots, x_k)$ in F, we add the constraint $C' = R'(\alpha, x_1, \ldots, x_k)$ to the S'-formula F', where α is a new variable. Note that F' is a S'-formula, defined on $n+1$ variables, which is satisfiable if and only if F is satisfiable. Moreover, the construction of F' from F can be done with a monotone projection. □

Since the case $\mathsf{Pol}(S) = \mathrm{L}_2$ holds, the case $\mathsf{Pol}(S) = \mathrm{L}_3$ now follows from Lemma 4.4 and the Claim. Finally, from Lemma 4.4 we conclude that the reduction also holds for the case $\mathsf{Pol}(S) \in \{\mathrm{N}_2, \mathrm{I}_2\}$, since $\mathrm{I}_2 \subseteq \mathrm{N}_2 \subseteq \mathrm{L}_3$. □

Theorem 4.8 (Dichotomy for monotone circuits)**.** *Let S be a finite set of relations. If $\mathsf{Pol}(S) \subseteq \mathrm{L}_3$ then there exists a constant $\varepsilon > 0$ such that $\mathsf{mSIZE}(\mathsf{CSP\text{-}SAT}_S) = 2^{\Omega(n^\varepsilon)}$. Otherwise, we have $\mathsf{mSIZE}(\mathsf{CSP\text{-}SAT}_S) = n^{O(1)}$.*

Proof. If $\mathsf{Pol}(S) \subseteq \mathrm{L}_3$, the lower bound follows from the 'moreover' part of Theorem 3.2, and Lemma 4.7. For the upper bound, we inspect Post's lattice (Figure 1). Observe that, if $\mathsf{Pol}(S) \not\subseteq \mathrm{L}_3$, the following are the only possible cases:

1. $\mathrm{I}_0 \subseteq \mathsf{Pol}(S)$ or $\mathrm{I}_1 \subseteq \mathsf{Pol}(S)$. In both cases, any $\mathsf{CNF}(S)$ is trivially satisfiable.
2. $\mathrm{E}_2 \subseteq \mathsf{Pol}(S)$ or $\mathrm{V}_2 \subseteq \mathsf{Pol}(S)$. In this case, $\mathsf{CSP\text{-}SAT}_S \in \mathsf{mSIZE}[\mathsf{poly}]$ by Theorem 4.5.
3. $\mathrm{D}_2 \subseteq \mathsf{Pol}(S)$. In this case, $\mathsf{CSP\text{-}SAT}_S \in \mathsf{mNL} \subseteq \mathsf{mSIZE}[\mathsf{poly}]$ by Theorem 4.5. □

Remark 4.9. *We remark that the lifting theorem of [GKRS19] (which is an ingredient in the proof of Theorem 3.2) is only used to prove that the monotone complexity of $\mathsf{CSP\text{-}SAT}_S$ is exponential when $\mathsf{Pol}(S) \subseteq \mathrm{L}_3$. If we only care to show a superpolynomial separation, then it suffices to apply the superpolynomial lower bound for CSPs with counting proved in [FV98, BGW99] using the approximation method. Indeed, we give an explicit proof in Section 5. The same holds for the consequences of this theorem (see Theorem 4.19).*

Dichotomy for formulas. Define $\mathsf{3\text{-}Horn\text{-}SAT}_n : \{0,1\}^{2n^3+n} \to \{0,1\}$ as $\mathsf{3\text{-}Horn\text{-}SAT}_n = \mathsf{CSP\text{-}SAT}^n_{\mathcal{H}^3}$, where

$$\mathcal{H}^3 = \{(\neg x_1 \vee \neg x_2 \vee x_3), (\neg x_1 \vee \neg x_2 \vee \neg x_3), (x)\}.$$

The following is proved in [RM99, GKRS19].

Theorem 4.10 ([RM99, GKRS19])**.** *There exists $\varepsilon > 0$ such that $\mathsf{3\text{-}Horn\text{-}SAT} \in \mathsf{mSIZE}[\mathsf{poly}] \setminus \mathsf{mDEPTH}[o(n^\varepsilon)]$.*

Proof sketch. Since $\mathrm{E}_2 \subseteq \mathsf{Pol}(\mathcal{H}^3)$ (see, e.g., [CKS01, Lemma 4.8]), the upper bound follows from Theorem 4.5. The lower bound follows from a lifting theorem of [RM99, GKRS19]. They show that the monotone circuit-depth of $\mathsf{3\text{-}Horn\text{-}SAT}$ is at least the depth of the smallest Resolution-tree refuting a so-called *pebbling formula*. Since this formula requires Resolution-trees of depth n^ε, the lower bound follows. □

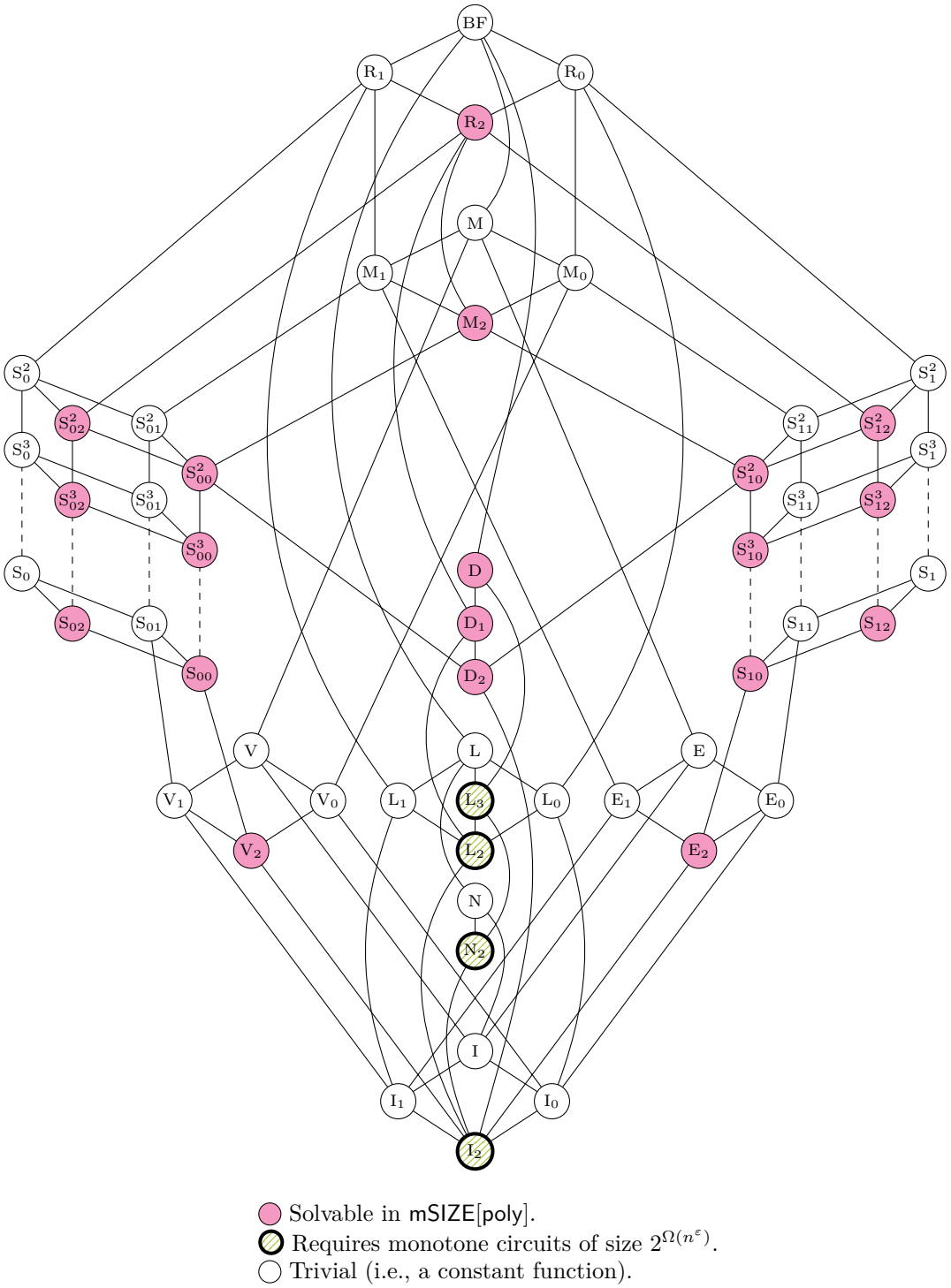


Figure 2: Illustration of Theorem 4.8. The vertices are colored with the *monotone circuit size* complexity of deciding CSPs whose set of polymorphisms corresponds to the label of the vertex.

Analogously to the previous section, we show that 3-Horn-SAT reduces to $\mathsf{CSP\text{-}SAT}_S$ whenever $\mathsf{Pol}(S)$ is small enough, in a precise sense stated below. We then deduce the dichotomy for formulas with a similar argument.

Lemma 4.11. *Let S be a finite set of relations. If $\mathsf{Pol}(S) \subseteq \mathrm{E}_2$ or $\mathsf{Pol}(S) \subseteq \mathrm{V}_2$, then 3-Horn-SAT $\leq_m^{\mathsf{mNL}}$ $\mathsf{CSP\text{-}SAT}_S$.*

Proof. We first consider the case $\mathsf{Pol}(S) \subseteq \mathrm{E}_2$. Note that $\mathrm{E}_2 \subseteq \mathsf{Pol}$(3-Horn-SAT) (see, e.g., [CKS01, Lemma 4.8]). Therefore, from Lemma 4.4 we deduce that 3-Horn-SAT admits a reduction to $\mathsf{CSP\text{-}SAT}_S$ in mNL.

Now let 3-AntiHorn-SAT $= \mathsf{CSP\text{-}SAT}_{\mathcal{A}^3}$, where $\mathcal{A}^3 = \{(x_1 \vee x_2 \vee \neg x_3), (x_1 \vee x_2 \vee x_3), (\neg x)\}$. Observe that a $\mathcal{H}^3$-formula φ is satisfiable if and only if the $\mathcal{A}^3$-formula $\varphi(\neg x_1, \ldots, \neg x_n)$ is satisfiable. Therefore, we have 3-Horn-SAT $\leq_m^{\mathsf{mProj}}$ 3-AntiHorn-SAT. Observing that $\mathrm{V}_2 \subseteq \mathsf{Pol}(\mathcal{A}^3)$ (again, see e.g. [CKS01, Lemma 4.8]), the result now follows from Lemma 4.4 and the previous paragraph. □

Theorem 4.12 (Dichotomy for monotone formulas)**.** *Let S be a finite set of relations. If $\mathsf{Pol}(S) \subseteq \mathrm{L}_3$, or $\mathsf{Pol}(S) \subseteq \mathrm{V}_2$, or $\mathsf{Pol}(S) \subseteq \mathrm{E}_2$, then there is a constant $\varepsilon > 0$ such that $\mathsf{mDEPTH}(\mathsf{CSP\text{-}SAT}_S) = \Omega(n^\varepsilon)$. Otherwise, we have $\mathsf{CSP\text{-}SAT}_S \in \mathsf{mNL} \subseteq \mathsf{mNC}^2 \subseteq \mathsf{mDEPTH}[\log^2 n]$.*

Proof. We will first prove the lower bound. If $\mathsf{Pol}(S) \subseteq \mathrm{L}_3$, the lower bound follows from Theorem 4.8. If $\mathsf{Pol}(S) \subseteq \mathrm{V}_2$ or $\mathsf{Pol}(S) \subseteq \mathrm{E}_2$, the lower bound follows from Theorem 4.10 and Lemma 4.11.

By inspecting Post's lattice (Remark 4.1), we see that the remaning cases are:

1. $\mathrm{I}_0 \subseteq \mathsf{Pol}(S)$ or $\mathrm{I}_1 \subseteq \mathsf{Pol}(S)$. In both cases, any $\mathsf{CNF}(S)$ is trivially satisfiable.
2. $\mathrm{S}_{00} \subseteq \mathsf{Pol}(S)$, or $\mathrm{S}_{10} \subseteq \mathsf{Pol}(S)$, or $\mathrm{D}_2 \subseteq \mathsf{Pol}(S)$. In all of those three cases, we have $\mathsf{CSP\text{-}SAT}_S \in$ mNL by Theorem 4.5. □

4.4 Some auxiliary results

In this section, we prove auxiliary results needed in the proof of a more general form of Theorem 3.5. In particular, we will prove that all $\mathsf{CSP\text{-}SAT}_S$ which are in AC^0 are also contained in $\mathsf{mAC}^0 \subseteq \mathsf{mNC}^1$. Moreover, we show that, if $\mathsf{CSP\text{-}SAT}_S \notin \mathsf{mNC}^1$, then $\mathsf{CSP\text{-}SAT}_S$ is L-hard under $\leq_m^{\mathsf{AC}^0}$ reductions.

We first observe that, when $\mathsf{COQ}(S_1) \subseteq \mathsf{COQ}(S_2)$, there exists an efficient low-depth reduction from $\mathsf{CSP\text{-}SAT}_{S_1}$ to $\mathsf{CSP\text{-}SAT}_{S_2}$. This reduction, which will be useful in this section, is more refined than the one given by Lemma 4.4. A proof of the non-monotone version of this statement is found in [BCRV04, Proposition 2.3], and we give a monotone version of this proof in Section 6.2.

Lemma 4.13 ([BCRV04, Proposition 2.3])**.** *If $\mathsf{COQ}(S_1) \subseteq \mathsf{COQ}(S_2)$, then there exists a constant $C \in \mathbb{N}$ such that $\mathsf{CSP\text{-}SAT}^n_{S_1} \leq_m^{\mathsf{mOR}} \mathsf{CSP\text{-}SAT}^{Cn}_{S_2}$.*

Proof. We defer the proof to Section 6.2. □

We now recall some lemmas from [ABI+09], and prove a few consequences from them. We say that a set S of relations *can express equality* if $\{=\} \subseteq \mathsf{COQ}(S)$.

Lemma 4.14 ([ABI+09])**.** *Let S be a finite set of relations. Suppose $\mathrm{S}_{02} \subseteq \mathsf{Pol}(S)$ ($\mathrm{S}_{12} \subseteq \mathsf{Pol}(S)$, resp.) and that S cannot express equality. Then there exists $k \geq 2$ such that $S \subseteq \mathsf{COQ}(\left\{\mathsf{OR}^k, x, \neg x\right\})$ ($S \subseteq \mathsf{COQ}(\left\{\mathsf{NAND}^k, x, \neg x\right\})$, resp.).*

Proof. Follows from the proof of Lemma 3.8 of [ABI+09]. □

Lemma 4.15. *Let S be a finite set of relations such that $\mathsf{Pol}(S) \subseteq \mathrm{R}_2$. If $\mathrm{S}_{02} \subseteq \mathsf{Pol}(S)$ or $\mathrm{S}_{12} \subseteq \mathsf{Pol}(S)$, and S cannot express equality, then $\mathsf{CSP\text{-}SAT}_S \in \mathsf{mAC}^0_3$.*

Proof. We write the proof in the case $\mathrm{S}_{02} \subseteq \mathsf{Pol}(S)$. The other case is analogous.

From Lemmas 4.13 and 4.14 and Items (iv) and (v) of Lemma 4.3, we get that there is a monotone OR-reduction from $\mathsf{CSP\text{-}SAT}_S$ to $\mathsf{CSP\text{-}SAT}_{\left\{\mathsf{OR}^k, x, \neg x\right\}}$ for some k. However, an $\left\{\mathsf{OR}^k, x, \neg x\right\}$-formula is

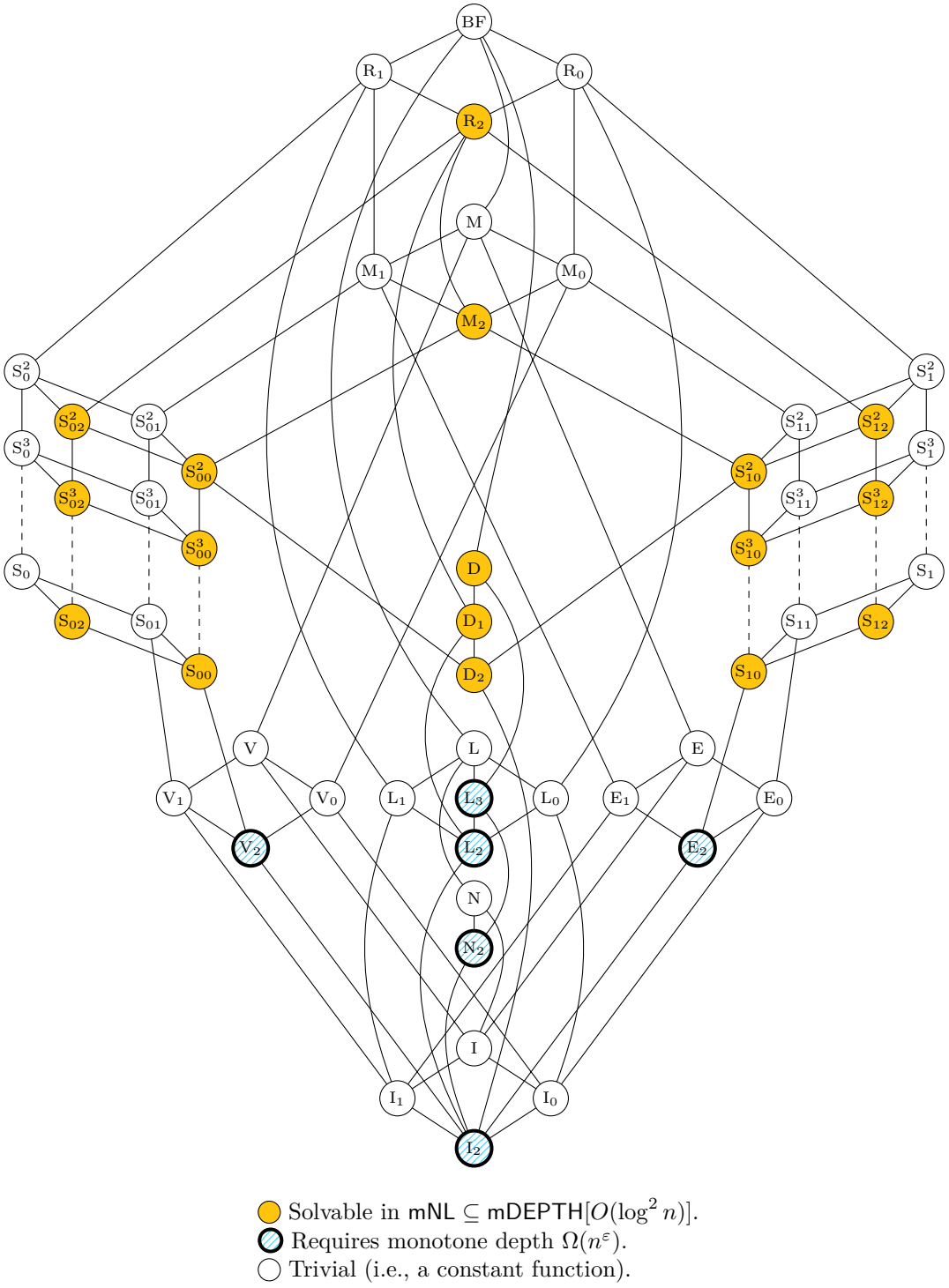


Figure 3: Illustration of Theorem 4.12. The vertices are colored with the *monotone circuit depth* complexity of deciding CSPs whose set of polymorphisms corresponds to the label of the vertex.

unsatisfiable iff there exists a literal and its negation as a constraint in the formula, or if there exists a disjunction in the formula such that every one of its literals appears negatively as a constraint. This condition can be easily checked by a polynomial-size monotone DNF. Composing the monotone DNF with the monotone OR-reduction, we obtain a depth-3 AC^0 circuit computing $\mathsf{CSP\text{-}SAT}_S$. □

Lemma 4.16 ([ABI+09, Lemma 3.8])**.** *Let S be a finite set of relations such that $\mathsf{Pol}(S) \subseteq \mathrm{R}_2$. If $\mathrm{S}_{02} \subseteq \mathsf{Pol}(S)$ or $\mathrm{S}_{12} \subseteq \mathsf{Pol}(S)$, and S can express equality, then $\mathsf{CSP\text{-}SAT}_S$ is L-hard under $\leq_m^{\mathsf{AC}^0}$ reductions.*

Lemma 4.17. *Let S be a finite set of relations. If $\mathrm{S}_{02} \not\subseteq \mathsf{Pol}(S)$ and $\mathrm{S}_{12} \not\subseteq \mathsf{Pol}(S)$, then $\mathsf{CSP\text{-}SAT}_S$ is L-hard or trivial.*

Proof. This follows by inspecting Post's lattice (Figure 1) and the classification theorem (Theorem 4.6). □

We may now prove the main result of this subsection.

Theorem 4.18. *We have $\mathsf{CSP} \cap \mathsf{AC}^0 \subseteq \mathsf{mAC}^0_3$. Moreover, if $\mathsf{CSP\text{-}SAT}_S \notin \mathsf{mAC}^0_3$, then $\mathsf{CSP\text{-}SAT}_S$ is L-hard under $\leq_m^{\mathsf{AC}^0}$ reductions.*

Proof. Let S be a finite set of relations. If $\mathsf{CSP\text{-}SAT}_S \notin \mathsf{mAC}^0_3$, then, by Lemma 4.15, at least one of the following cases hold:

1. $\mathrm{S}_{02} \subseteq \mathsf{Pol}(S) \subseteq \mathrm{R}_2$ or $\mathrm{S}_{12} \subseteq \mathsf{Pol}(S) \subseteq \mathrm{R}_2$, and S can express the equality relation;
2. $\mathrm{S}_{02} \not\subseteq \mathsf{Pol}(S) \subseteq \mathrm{R}_2$ and $\mathrm{S}_{12} \not\subseteq \mathsf{Pol}(S) \subseteq \mathrm{R}_2$.
3. $\mathsf{Pol}(S) \not\subseteq \mathrm{R}_2$.

Since $\mathsf{CSP\text{-}SAT}_S$ is not trivial, we obtain that $\mathsf{CSP\text{-}SAT}_S$ is L-hard in the first two cases by Lemmas 4.16 and 4.17, and it's easy to check that $\mathsf{CSP\text{-}SAT}_S$ is also L-hard in the third case by inspecting Post's lattice (Figure 1) and the classification theorem (Theorem 4.6). Since $\mathsf{L} \not\subseteq \mathsf{AC}^0$, this also implies that, if $\mathsf{CSP\text{-}SAT}_S \in \mathsf{AC}^0$, then $\mathrm{S}_{02} \subseteq \mathsf{Pol}(S) \subseteq \mathrm{R}_2$ or $\mathrm{S}_{12} \subseteq \mathsf{Pol}(S) \subseteq \mathrm{R}_2$, and S cannot express the equality relation. Lemma 4.15 again gives $\mathsf{CSP\text{-}SAT}_S \in \mathsf{mAC}^0_3$. □

4.5 Consequences for monotone circuit lower bounds via lifting

We now prove a stronger form of Theorem 3.5. In the previous section, we showed that $\mathsf{CSP} \cap \mathsf{AC}^0 \subseteq \mathsf{mAC}^0$. In particular, this means that there does not exist a finite set of relations S such that $\mathsf{CSP\text{-}SAT}_S$ separates AC^0 and mNC^1, a separation which we proved in Theorem 3.3. We will also observe that, if $\mathsf{CSP\text{-}SAT}_S \notin \mathsf{mNC}^2$, then $\mathsf{CSP\text{-}SAT}_S$ is $\oplus\mathsf{L}$-hard.

Theorem 4.19. *Let S be a finite set of Boolean relations.*

1. *If $\mathsf{CSP\text{-}SAT}_S \notin \mathsf{mAC}^0_3$ then $\mathsf{CSP\text{-}SAT}_S$ is L-hard under $\leq_m^{\mathsf{AC}^0}$ reductions.*
2. *If $\mathsf{CSP\text{-}SAT}_S \notin \mathsf{mNC}^2$, then $\mathsf{CSP\text{-}SAT}_S$ is $\oplus\mathsf{L}$-hard under $\leq_m^{\mathsf{AC}^0}$ reductions.*

Proof. Item (1) follows from Theorem 4.18. To prove item (2), suppose that $\mathsf{mDEPTH}(\mathsf{CSP\text{-}SAT}_S) = \omega(\log^2 n)$. Then, by Theorem 4.12, we conclude that $\mathsf{Pol}(S) \subseteq \mathrm{L}_3$, or $\mathsf{Pol}(S) \subseteq \mathrm{V}_2$, or $\mathsf{Pol}(S) \subseteq \mathrm{E}_2$. Theorem 4.6 implies that $\mathsf{CSP\text{-}SAT}_S$ is $\oplus\mathsf{L}$-hard. □

Further Discussion. We recall the discussion of Section 3.2. We introduced and defined the functions $\mathsf{CSP\text{-}SAT}_S$ in that section, as a way to capture monotone circuit lower bounds proved via lifting. This in particular captures the monotone function 3-XOR-SAT, which was proved in [GKRS19] to require monotone circuit lower bounds of size $2^{n^{\Omega(1)}}$ to compute, even though $\oplus\mathsf{L}$-machines running in polynomial-time can compute it. Theorem 4.19 proves that this separation between monotone and non-monotone circuit lower bounds cannot be improved by varying the set of relations S, as we argue below.

There are two ways one could try to find a function in AC^0 with large monotone complexity using a CSP-SAT function. First, one could try to define a set of relations S such that $\mathsf{CSP\text{-}SAT}_S \in \mathsf{AC}^0$, but the monotone complexity of $\mathsf{CSP\text{-}SAT}_S$ is large. However, Item (1) of Theorem 4.19 proves that this is

impossible, as any CSP-SAT function outside of mAC^0 is L-hard under simple reductions and, therefore, cannot be computed in AC^0.

Secondly, one could try to be apply the arguments of Original reference 3, consisting of a padding trick and a simulation theorem. When S is the set of 3XOR relations, then indeed we obtain a function in $\mathsf{AC}^0[\oplus]$ with superpolynomial monotone circuit complexity, as proved in Theorem 3.2. However, Item (2) of Theorem 4.19 proves that this is best possible, as any CSP-SAT function which admits a superpolynomial monotone circuit lower bound must be $\oplus$L-hard and, therefore, at least as hard as 3-XOR-SAT for non-monotone circuits. Item (2) also shows that even CSP-SAT functions with a $\omega(\log^2 n)$ monotone depth lower bound must be $\oplus$L-hard, which suggests that the arguments of Original reference 3 applied to a CSP-SAT function are not able to prove the separation of Theorem 3.6.

A caveat to these impossibility results is in order. First, we only study Boolean-valued CSPs here, though the framework of lifting can also be applied in the context of non-Boolean CSPs. It's not clear if non-Boolean CSPs exhibit the same dichotomies for monotone computation we proved in this section. We remark that Schaefer's dichotomy for Boolean-valued CSPs [Sch78] has been extended to non-Boolean CSPs [Zhu17, Bul17].

Secondly, the instances of CSP-SAT generated by lifting do not cover the entirety of the minterms and maxterms of CSP-SAT. In particular, our results do not rule out the possibility that a clever interpolation of the instances generated by lifting may give rise to a function that is easier to compute by non-monotone circuits, and therefore bypasses the hardness results of Theorem 4.19. One example is the Tardós function [Tar88]. A lifting theorem applied to a Pigeonhole Principle formula can be used to prove a lower bound on the size of monotone circuits that accept cliques of size k and reject graphs that are $(k-1)$-colorable, for some $k = n^{\varepsilon}$ [RM99, Rez23]. A natural interpolation for these instances would be the k-Clique function, which, being NP-complete, would be related to an NP-complete CSP-SAT. However, as proved by [Tar88], there is a monotone function in P which has the same output behaviour over these instances.

5 A Lower Bound for 3-XOR-SAT Using the Approximation Method

As discussed in Section 3.2, [GKRS19] obtained an exponential lower bound on the monotone circuit size of the function 3-XOR-SAT using techniques from communication complexity and lifting. Here we observe that a weaker but still super-polynomial lower bound can be proved using the approximation method.

First, we recall the function $\mathsf{OddFactor}_n : \{0,1\}^{\binom{n}{2}} \to \{0,1\}$ of Original reference 3.2, which accepts a given graph if the graph contains an *odd factor*, which is a spanning subgraph in which the degree of every vertex is odd. For convenience, in this section we consider a weaker version of OddFactor, which takes as an input a *bipartite* graph with n vertices on each part, and accepts if the graph contains an odd factor. Let $\mathsf{Bipartite\text{-}OddFactor}_n : \{0,1\}^{n^2} \to \{0,1\}$ be this function. We remark that the lower bounds of Babai, Gál and Wigderson [BGW99] for OddFactor (Theorem 3.7) also hold for Bipartite-OddFactor. The proof of the monotone circuit lower bound in particular is essentially Razborov's lower bound for Matching via the approximation method [Raz85b].

Theorem 5.1 ([BGW99])**.** *We have*

$$\mathsf{mSIZE}(\mathsf{Bipartite\text{-}OddFactor}_n) = n^{\Omega(\log n)} \quad \textit{and} \quad \mathsf{mDEPTH}(\mathsf{Bipartite\text{-}OddFactor}_n) = \Omega(n).$$

We can reduce Bipartite-OddFactor to 3-XOR-SAT by noting that computing $\mathsf{Bipartite\text{-}OddFactor}_n(M)$ on a given matrix $M \in \{0,1\}^{n^2}$ is computationally equivalent to deciding the satisfiability of the following $\mathbb{F}_2$ linear system over variables $\{x_{ij}\}$:

- For all $i \in [n]$: $\bigoplus_{k=1}^{n} x_{ik} = 1$;
- For all $j \in [n]$: $\bigoplus_{k=1}^{n} x_{kj} = 1$;
- For all $i, j \in [n]$ such that $M_{ij} = 0$: $x_{ij} = 0$.

We can then use a circuit for 3-XOR-SAT to solve this system by using a standard trick of introducing new variables to reduce the number of variables that appear in each equation, as done in the textbook

reduction from SAT to 3-SAT. As the corresponding reductions turn out to be monotone, this implies monotone circuit and formula lower bounds for 3-XOR-SAT. We note that a somewhat similar argument (in the non-monotone setting) appears in Feder and Vardi [FV98, Theorem 30] regarding constraint satisfaction problems with the ability to count.

In order to formalise this argument, we will need the following definition and results.

Definition 5.2. *Let f be a Boolean function. We define* $\mathsf{dual}(f) : x \mapsto \neg f(\neg x)$ *as the* dual *of f.*

Lemma 5.3. *Let f be a monotone Boolean function. We have* $\mathsf{mSIZE}(f) = \mathsf{mSIZE}(\mathsf{dual}(f))$ *and* $\mathsf{mDEPTH}(f) = \mathsf{mDEPTH}(\mathsf{dual}(f))$.

Proof. The idea is to push negations to the bottom and eliminate double negations at the input layer. In other words, applying De Morgan rules, we can convert any $\{\wedge, \vee\}$-circuit computing f into a circuit computing $\mathsf{dual}(f)$ by swapping $\wedge$-gates for $\vee$-gates, and vice-versa. Moreover, this transformation preserves the depth of the circuit. □

We are ready to describe a monotone reduction from the function $\mathsf{Bipartite\text{-}OddFactor}_n$ to 3-XOR-SAT, which implies the desired lower bounds.

Theorem 5.4. *There exists $\varepsilon > 0$ such that*

$$\mathsf{mSIZE}(\text{3-XOR-SAT}) = n^{\Omega(\log n)} \quad \textit{and} \quad \mathsf{mDEPTH}(\text{3-XOR-SAT}) = \Omega(n^{\varepsilon}).$$

Proof. Recall that the value of the function $\mathsf{Bipartite\text{-}OddFactor}_n(M)$ on a given matrix $M \in \{0,1\}^{n^2}$ is equal to 1 if the following system is satisfiable:

- For all $i \in [n]$: $\bigoplus_{k=1}^{n} x_{ik} = 1$;
- For all $j \in [n]$: $\bigoplus_{k=1}^{n} x_{kj} = 1$;
- For all $i, j \in [n]$ such that $M_{ij} = 0$: $x_{ij} = 0$.

We introduce some extra variables to reduce the number of variables in each equation in the following way. For every $i \in [n]$, introduce variables $z_{i1}, \ldots, z_{i(n-1)}$ and the equations

$$\begin{aligned} z_{i1} &= x_{i1} \oplus x_{i2}, \\ z_{i2} &= z_{i1} \oplus x_{i3}, \\ &\ldots \\ z_{i,(n-1)} &= z_{i,(n-2)} \oplus x_{i,n}, \\ z_{i,(n-1)} &= 1. \end{aligned}$$

Now note that these equations imply $z_{i,(n-2)} = \bigoplus_{k=1}^{n} x_{ik} = 1$. For each "column" equation $\bigoplus_{k=1}^{n} x_{kj} = 1$, we also add variables $w_{j1}, \ldots, w_{j(n-1)}$ as above. In total, we add at most $2n^2$ variables and $2n^2$ equations. Therefore, there is a system of linear equations on $O(n^2)$ variables, where each constraint contains at most 3 variables, which is satisfiable if and only if $\mathsf{Bipartite\text{-}OddFactor}_n(M) = 1$. Moreover, it is easy to see that the characteristic vector α of the set of equations of this system can be computed from M by an *anti-monotone* projection, as we activate a constraint that depends on the input when $M_{ij} = 0$.

Now let $f = \mathsf{dual}(\text{3-XOR-SAT})$ and $\beta = \neg\alpha$. Since, by definition, 3-XOR-SAT accepts *unsatisfiable* systems, we get $\mathsf{Bipartite\text{-}OddFactor}_n(M) = \neg\text{3-XOR-SAT}(\alpha) = f(\beta)$ and that β is a *monotone* projection of M. Therefore, by Lemma 5.3, we obtain

$$\mathsf{mSIZE}(\mathsf{Bipartite\text{-}OddFactor}_n) \leq \mathsf{mSIZE}(\text{3-XOR-SAT})$$

and

$$\mathsf{mDEPTH}(\mathsf{Bipartite\text{-}OddFactor}_n) \leq \mathsf{mDEPTH}(\text{3-XOR-SAT}).$$ □

6 Schaefer's Theorem in Monotone Complexity

6.1 Connectivity and generation functions

We recall the definitions of two prominent monotone Boolean functions that have efficient monotone circuits. Let $\mathsf{ST\text{-}CONN} : \{0,1\}^{n^2} \to \{0,1\}$ be the function that outputs 1 on a given directed graph G if there exists a path from 1 to n in G. Let $\mathsf{GEN} : \{0,1\}^{n^3} \to \{0,1\}$ be the Boolean function which receives a set T of triples $(i,j,k) \in [n^3]$, and outputs 1 if $n \in S$, where $S \subseteq [n]$ is the set generated with the following rules:

- *Axiom:* $1 \in S$,
- *Generation:* If $i, j \in S$ and $(i,j,k) \in T$, then $k \in S$.

The following upper bounds are well-known and easy to prove.

Theorem 6.1 ([Juk12, Exercise 7.3], [RM99])**.** *We have* $\mathsf{ST\text{-}CONN} \in \mathsf{mNL}$ *and* $\mathsf{GEN} \in \mathsf{mSIZE[poly]}$.

6.2 Proof of reduction lemmas

Here we present monotonised versions of the proofs of [BCRV04, Propositions 2.2 - 2.4], which give a simplified presentation of the results of [Sch78].

Lemma 4.13 ([BCRV04, Proposition 2.3])**.** *If* $\mathsf{COQ}(S_1) \subseteq \mathsf{COQ}(S_2)$*, then there exists a constant* $C \in \mathbb{N}$ *such that* $\mathsf{CSP\text{-}SAT}^n_{S_1} \leq^{\mathsf{mOR}}_m \mathsf{CSP\text{-}SAT}^{Cn}_{S_2}$.

Proof. If $\mathsf{COQ}(S_1) \subseteq \mathsf{COQ}(S_2)$, then each relation of S_1 can be represented as a conjunctive query over S_2. Let F_1 be a S_1-formula. For each constraint C_1 of F_1, there exists a formula $\varphi(C_1)$ in $\mathsf{CNF}(S_2)$ such that C_1 is a projection of $\varphi(C_1)$ (i.e., C_1 is a conjunctive query of $\varphi(C_1)$). However, note that C_1 is satisfiable if and only if $\varphi(C_1)$ is satisfiable. So we can replace the constraint C_1 by the *set* of constraints in $\varphi(C_1)$. Doing this for every constraint in F_1, we obtain an S_2-formula F_2 which is satisfiable iff F_1 is satisfiable.

Now note that, to decide if a given constraint application C of S_2 is in the reduction, it suffices to check if there exists a S_1-constraint C_1 in F_1 such that C is in $\varphi(C_1)$. Using non-uniformity, this can be easily done by an OR over the relevant input bits.

Finally, we observe that, since the arities of each relation in S_1 and S_2 are constant, we only add a constant number of variables for each constraint to represent S_1-formulas with conjunctive queries over S_2-formulas. □

Lemma 6.2. *Let* S *be a set of Boolean relations. We have* $\mathsf{CSP\text{-}SAT}_{S\cup\{=\}} \leq^{\mathsf{mNL}}_m \mathsf{CSP\text{-}SAT}_S$.

Proof. Let F be a $(S \cup \{=\})$-formula on n variables given as an input. Remember that F is given as a Boolean vector α, where each bit of α represents the presence of a constraint application on n variables from $S \cup \{=\}$. We first build an undirected graph G with the variables $x_1, \ldots, x_n$ as vertices, and we put an edge between x_i and x_j if the constraint $x_i = x_j$ appears in F. Note that G can be constructed by a monotone projection from F.

Let $R \in S$ and let $C = R(x_1, \ldots, x_n)$ be a constraint application of R. If C appears in F, we add to the system every constraint of the form $C' = R(y_1, \ldots, y_n)$ such that, for every $i \in [n]$, there exists a path from x_i to y_i in the graph G. In this case, we say that C *generates* C'. Let F_2 be the formula that contains all non-equality constraints of F, and all the non-equality constraints generated by a constraint in F. It's not hard to see that F is satisfiable if and only if F_2 is satisfiable, and therefore the reduction is correct.

Moreover, the reduction can be done in monotone NL using the monotone NL algorithm for $\mathsf{ST\text{-}CONN}$ (Theorem 6.1). Indeed, there are at most n^k constraint applications of a given relation R of arity k. Therefore, to decide if a constraint $C' = R(y_1, \ldots, y_n)$ appears in F_2, it suffices to check if there exists a constraint application of R in F which generates C'. This can be checked with n^k calls to $\mathsf{ST\text{-}CONN}$. □

Lemma 4.4 (Polymorphisms characterise the complexity of CSPs [Jea98, BCRV04, Theorem 2.4])**.** *If* $\mathsf{Pol}(S_2) \subseteq \mathsf{Pol}(S_1)$*, then* $\mathsf{CSP\text{-}SAT}^n_{S_1} \leq^{\mathsf{mNL}}_m \mathsf{CSP\text{-}SAT}^{\mathsf{poly}(n)}_{S_2}$.

Proof. If $\mathsf{Pol}(S_2) \subseteq \mathsf{Pol}(S_1)$, then from Lemma 4.3 (Items (iii), (v) and (vi)) we obtain $\mathsf{COQ}(S_1) \subseteq \langle S_1 \rangle \subseteq \langle S_2 \rangle = \mathsf{COQ}(S_2 \cup \{=\})$. Therefore, by Lemmas 4.13 and 6.2 we can do the following chain of reductions in monotone NL:

$$\mathsf{CSP\text{-}SAT}_{S_1} \leq_m^{\mathsf{mOR}} \mathsf{CSP\text{-}SAT}_{S_2 \cup \{=\}} \leq_m^{\mathsf{mNL}} \mathsf{CSP\text{-}SAT}_{S_2}. \qquad \square$$

6.3 Monotone circuit upper bounds

We restate and prove the theorem.

Theorem 4.5 (Monotone version of the upper bounds for CSP-SAT [Sch78, ABI$^+$09])**.** *Let S be a finite set of relations. The following holds.*

1. *If* $\mathrm{E}_2 \subseteq \mathsf{Pol}(S)$ *or* $\mathrm{V}_2 \subseteq \mathsf{Pol}(S)$*, then* $\mathsf{CSP\text{-}SAT}_S \in \mathsf{mSIZE}[\mathsf{poly}]$.
2. *If* $\mathrm{D}_2 \subseteq \mathsf{Pol}(S)$*, or* $\mathrm{S}_{00} \subseteq \mathsf{Pol}(S)$*, or* $\mathrm{S}_{10} \subseteq \mathsf{Pol}(S)$*, then* $\mathsf{CSP\text{-}SAT}_S \in \mathsf{mNL}$.

Proof. We prove each case separately.

Proof of (1). We first observe that 3-Horn-SAT (see definition in Section 4.3, *Dichotomy for formulas*) can be solved by a reduction to $\mathsf{GEN} \in \mathsf{mSIZE}[\mathsf{poly}]$. Indeed, we interpret each constraint of the form $(\neg x_i \vee \neg x_j \vee x_k)$ (which is equivalent to $x_i \wedge x_j \implies x_k$) as a triple (i, j, k), and constraints of the form x_i as a triple $(0, 0, i)$. Let $S \subseteq \{0, 1, 2 \ldots, n\}$ be the set generated by these triples, applying the generation rules of GEN, using $0 \in S$ as the axiom. It suffices to check that there exists some constraint of the form $\neg x_i \vee \neg x_j \vee \neg x_k$, such that $\{i, j, k\} \subseteq S$. This process can be done with polynomial-size monotone circuits, invoking GEN. Therefore, it follows from Theorem 6.1 that $\text{3-Horn-SAT} \in \mathsf{mSIZE}[\mathsf{poly}]$.

Moreover, we recall that, if $\mathrm{E}_2 \subseteq \mathsf{Pol}(S)$, then $S \subseteq \mathsf{COQ}(\mathcal{H}^3)$ (in other words, every S-formula can be written as a set of 3-Horn equations) – see, e.g, [CKS01, Lemma 4.8]. Therefore, from Items (iv) and (v) of Lemma 4.3 and Lemma 4.13, we conclude that $\mathsf{CSP\text{-}SAT}_S \leq_m^{\mathsf{mOR}} \text{3-Horn-SAT} \in \mathsf{mSIZE}[\mathsf{poly}]$.

Now recall that, if $\mathrm{V}_2 \subseteq \mathsf{Pol}(S)$, then $S \subseteq \mathsf{COQ}(\mathcal{A}^3)$, where $\mathcal{A}^3$ is the set of width-3 Anti-Horn relations (i.e., $\mathcal{A}^3 = \{(x_1 \vee x_2 \vee \neg x_3), (x_1 \vee x_2 \vee x_3), (\neg x)\}$; see [CKS01, Lemma 4.8] for a proof of this observation). But note that an $\mathcal{A}^3$-formula φ is satisfiable if and only if the $\mathcal{H}^3$-formula $\varphi(\neg x_1, \ldots, \neg x_n)$ is satisfiable. Therefore by Lemma 4.13 and Items (iv) and (v) of Lemma 4.3, we have $\mathsf{CSP\text{-}SAT}_S \leq_m^{\mathsf{mOR}} \mathsf{CSP\text{-}SAT}_{\mathcal{A}^3} \leq_m^{\mathsf{mProj}} \text{3-Horn-SAT} \in \mathsf{mSIZE}[\mathsf{poly}]$.

Proof of (2). We first prove the case $\mathrm{D}_2 \subseteq \mathsf{Pol}(S)$. Let $\text{2-SAT} = \mathsf{CSP\text{-}SAT}_\Gamma$, where

$$\Gamma = \{(x_1 \vee x_2), (x_1 \vee \neg x_2), (\neg x_1 \vee \neg x_2)\}$$

. It's easy to check that the standard reduction from 2-SAT to ST-CONN can be done in monotone NL (see [JLL76, Theorem 4]). Therefore, it follows from Theorem 6.1 that $\text{2-SAT} \in \mathsf{mNL}$. Now, recall that, if $\mathrm{D}_2 \subseteq \mathsf{Pol}(S)$, then $S \subseteq \mathsf{COQ}(\Gamma)$ (see, e.g., [CKS01, Lemma 4.9]). Therefore, from Lemma 4.13 and Items (iv) and (v) of Lemma 4.3, we conclude $\mathsf{CSP\text{-}SAT}_S \in \mathsf{mNL}$.

We now suppose that $\mathrm{S}_{00} \subseteq \mathsf{Pol}(S)$. We check that the proof of [ABI$^+$09, Lemma 3.4] gives a monotone circuit. If $\mathrm{S}_{00} \subseteq \mathsf{Pol}(S)$, then there exists $k \geq 2$ such that ${\mathrm{S}_{00}}^k \subseteq \mathsf{Pol}(S)$ (that's because there does not exist a finite set of relations S such that $\mathsf{Pol}(S) = \mathrm{S}_{00}$). Note that ${\mathrm{S}_{00}}^k = \mathsf{Pol}(\Gamma)$, where $\Gamma = \left\{\mathsf{OR}^k, x, \neg x, \rightarrow, =\right\}$. We show below how to decide if a Γ-formula is unsatisfiable in monotone NL. The result then follows from Lemma 4.4.

Let F be a given Γ-formula with n variables. We first construct a directed graph G, with vertex set $\{x_1, \ldots, x_n\}$, and with arcs (x_i, x_j) if $x_i \to x_j$ is a constraint of F, and arcs (x_i, x_j) and (x_j, x_i) if $x_i = x_j$ is a constraint of F. This can be done with a monotone projection. Observe that a Γ-formula F is unsatisfiable if, and only if, there exists a constraint of the form $x_{i_1} \vee \cdots \vee x_{i_k}$ in F, such that there exists a path from some x_{i_j} to a constraint $\neg y$ in F. This can be checked in monotone NL by Theorem 6.1.

The case $\mathrm{S}_{10} \subseteq \mathsf{Pol}(S)$ is analogous. $\square$

7 Background on Post's Lattice and Clones

In this section, we include the definitions of the various clones that are used in the paper, as well as a figure of Post's lattice, which can be helpful when checking the proofs of Section 4.

Let $\rightarrow\colon (x,y) \mapsto (\neg x \vee y)$. Let also $\leftrightarrow\colon (x,y) \mapsto \neg(x \oplus y)$ and $\mathrm{id} : x \mapsto x$. Let $f : \{0,1\}^k \to \{0,1\}$ be a Boolean function. We say that f is *linear* if there exists $c \in \{0,1\}^k$ and $b \in \{0,1\}$ such that $f(x) = \langle c, x\rangle + b \pmod 2$. We say that f is self-dual if $f = \mathsf{dual}(f)$. Let $a \in \{0,1\}$. We say that f is *a-reproducing* if $f(a,\ldots,a) = a$. We say that a set $T \subseteq \{0,1\}^k$ is *a-separating* if there exists $i \in [k]$ such that $x_i = a$ for all $x \in T$. We say that f is *a-separating* if $f^{-1}(a)$ is a-separating. We say that f is *a-separating of degree k* if every $T \subseteq f^{-1}(a)$ such that $|T| = k$ is a-separating. The *basis* of a clone B is a set of Boolean functions F such that $B = [F]$.

References

[AB87] Noga Alon and Ravi B. Boppana. The monotone circuit complexity of boolean functions. *Combinatorica*, 7(1):1–22, 1987. 2

[ABI+09] Eric Allender, Michael Bauland, Neil Immerman, Henning Schnoor, and Heribert Vollmer. The complexity of satisfiability problems: Refining Schaefer's theorem. *J. Comput. Syst. Sci.*, 75(4):245–254, 2009. 3, 5, 6, 7, 9, 11, 15, 17

[AG87] Miklós Ajtai and Yuri Gurevich. Monotone versus positive. *J. Assoc. Comput. Mach.*, 34(4):1004–1015, 1987.

[AHM+08] Eric Allender, Lisa Hellerstein, Paul McCabe, Toniann Pitassi, and Michael E. Saks. Minimizing disjunctive normal form formulas and AC^0 circuits given a truth table. *SIAM J. Comput.*, 38(1):63–84, 2008. 2

[AK11] Jin Akiyama and Mikio Kano. *Factors and Factorizations of Graphs*, volume 2031 of *Lecture Notes in Mathematics*. Springer Berlin Heidelberg, Berlin, Heidelberg, 2011.

[AKR+01] Eric Allender, Michal Koucký, Detlef Ronneburger, Sambuddha Roy, and V. Vinay. Time-space tradeoffs in the counting hierarchy. In *Proceedings of the 16th Annual IEEE Conference on Computational Complexity, Chicago, Illinois, USA, June 18-21, 2001*, pages 295–302. IEEE Computer Society, 2001.

[AKS83] Miklós Ajtai, János Komlós, and Endre Szemerédi. An $O(n \log n)$ sorting network. In *Proceedings of the 15th Annual ACM Symposium on Theory of Computing, 25-27 April, 1983, Boston, Massachusetts, USA*, pages 1–9. ACM, 1983.

[And85] Alexander E Andreev. On a method for obtaining lower bounds for the complexity of individual monotone functions. *Doklady Akademii Nauk SSSR*, 282:1033–1037, 1985. 2

[BCRV03] Elmar Böhler, Nadia Creignou, Steffen Reith, and Heribert Vollmer. Playing with boolean blocks, part i: Post's lattice with applications to complexity theory. *SIGACT News*, 2003. 3, 4, 5

[BCRV04] Elmar Böhler, Nadia Creignou, Steffen Reith, and Heribert Vollmer. Playing with boolean blocks, part ii: Constraint satisfaction problems. *ACM SIGACT-Newsletter*, 35, 2004. 3, 5, 9, 14

[BDHM92] Gerhard Buntrock, Carsten Damm, Ulrich Hertrampf, and Christoph Meinel. Structure and importance of logspace-mod class. *Math. Syst. Theory*, 25(3):223–237, 1992.

[Ber82] S. J. Berkowitz. On some relationships between monotone and non-monotone circuit complexity. Technical Report. University of Toronto, 1982.

[BGW99] László Babai, Anna Gál, and Avi Wigderson. Superpolynomial lower bounds for monotone span programs. *Combinatorica*, 19(3):301–319, 1999. 3, 7, 12

[BHST14] Eric Blais, Johan Håstad, Rocco A. Servedio, and Li-Yang Tan. On DNF approximators for monotone boolean functions. In *International Colloquium on Automata, Languages, and Programming* (ICALP), pages 235–246, 2014.

Name	Definition	Base
BF	All Boolean functions	$\{\vee, \wedge, \neg\}$
R_0	$\{f \in \mathrm{BF} : f \text{ is 0-reproducing}\}$	$\{\wedge, \oplus\}$
R_1	$\{f \in \mathrm{BF} : f \text{ is 1-reproducing}\}$	$\{\vee, \leftrightarrow\}$
R_2	$\mathrm{R}_1 \cap \mathrm{R}_0$	$\{\vee, x \wedge (y \leftrightarrow z)\}$
M	$\{f \in \mathrm{BF} : f \text{ is monotonic}\}$	$\{\vee, \wedge, 0, 1\}$
M_1	$\mathrm{M} \cap \mathrm{R}_1$	$\{\vee, \wedge, 1\}$
M_0	$\mathrm{M} \cap \mathrm{R}_0$	$\{\vee, \wedge, 0\}$
M_2	$\mathrm{M} \cap \mathrm{R}_2$	$\{\vee, \wedge\}$
$\mathrm{S}_0^{\mathrm{n}}$	$\{f \in \mathrm{BF} : f \text{ is 0-separating of degree } n\}$	$\{\rightarrow, \mathsf{dual}(h_n)\}$
S_0	$\{f \in \mathrm{BF} : f \text{ is 0-separating}\}$	$\{\rightarrow\}$
$\mathrm{S}_1^{\mathrm{n}}$	$\{f \in \mathrm{BF} : f \text{ is 1-separating of degree } n\}$	$\{x \wedge \overline{y}, h_n\}$
S_1	$\{f \in \mathrm{BF} : f \text{ is 1-separating}\}$	$\{x \wedge \overline{y}\}$
$\mathrm{S}_{02}^{\mathrm{n}}$	$\mathrm{S}_0^{\mathrm{n}} \cap \mathrm{R}_2$	$\{x \vee (y \wedge \overline{z}), \mathsf{dual}(h_n)\}$
S_{02}	$\mathrm{S}_0 \cap \mathrm{R}_2$	$\{x \vee (y \wedge \overline{z})\}$
$\mathrm{S}_{01}^{\mathrm{n}}$	$\mathrm{S}_0^{\mathrm{n}} \cap \mathrm{M}$	$\{\mathsf{dual}(h_n), 1\}$
S_{01}	$\mathrm{S}_0 \cap \mathrm{M}$	$\{x \vee (y \wedge z), 1\}$
$\mathrm{S}_{00}^{\mathrm{n}}$	$\mathrm{S}_0^{\mathrm{n}} \cap \mathrm{R}_2 \cap \mathrm{M}$	$\{x \vee (y \wedge z), \mathsf{dual}(h_n)\}$
S_{00}	$\mathrm{S}_0 \cap \mathrm{R}_2 \cap \mathrm{M}$	$\{x \vee (y \wedge z)\}$
$\mathrm{S}_{12}^{\mathrm{n}}$	$\mathrm{S}_1^{\mathrm{n}} \cap \mathrm{R}_2$	$\{x \wedge (y \vee \overline{z}), h_n\}$
S_{12}	$\mathrm{S}_1 \cap \mathrm{R}_2$	$\{x \wedge (y \vee \overline{z})\}$
$\mathrm{S}_{11}^{\mathrm{n}}$	$\mathrm{S}_1^{\mathrm{n}} \cap \mathrm{M}$	$\{h_n, 0\}$
S_{11}	$\mathrm{S}_1 \cap \mathrm{M}$	$\{x \wedge (y \vee z), 0\}$
$\mathrm{S}_{10}^{\mathrm{n}}$	$\mathrm{S}_1^{\mathrm{n}} \cap \mathrm{R}_2 \cap \mathrm{M}$	$\{x \wedge (y \vee z), h_n\}$
S_{10}	$\mathrm{S}_1 \cap \mathrm{R}_2 \cap \mathrm{M}$	$\{x \wedge (y \vee z)\}$
D	$\{f \in \mathrm{BF} : f \text{ is self-dual}\}$	$\{(x \wedge \overline{y}) \vee (x \wedge \overline{z}) \vee (\overline{y} \wedge \overline{z})\}$
D_1	$\mathrm{D} \cap \mathrm{R}_2$	$\{(x \wedge y) \vee (x \wedge \overline{z}) \vee (y \wedge \overline{z})\}$
D_2	$\mathrm{D} \cap \mathrm{M}$	$\{(x \wedge y) \vee (y \wedge z) \vee (x \wedge z)\}$
L	$\{f \in \mathrm{BF} : f \text{ is linear}\}$	$\{\oplus, 1\}$
L_0	$\mathrm{L} \cap \mathrm{R}_0$	$\{\oplus\}$
L_1	$\mathrm{L} \cap \mathrm{R}_1$	$\{\leftrightarrow\}$
L_2	$\mathrm{L} \cap \mathrm{R}$	$\{x \oplus y \oplus z\}$
L_3	$\mathrm{L} \cap \mathrm{D}$	$\{x \oplus y \oplus z \oplus 1\}$
V	$\{f \in \mathrm{BF} : f \text{ is constant or an } n\text{-ary OR function}\}$	$\{\vee, 0, 1\}$
V_0	$[\{\vee\}] \cup [\{0\}]$	$\{\vee, 0\}$
V_1	$[\{\vee\}] \cup [\{1\}]$	$\{\vee, 1\}$
V_2	$[\{\vee\}]$	$\{\vee\}$
E	$\{f \in \mathrm{BF} : f \text{ is constant or an } n\text{-ary AND function}\}$	$\{\wedge, 0, 1\}$
E_0	$[\{\wedge\}] \cup [\{0\}]$	$\{\wedge, 0\}$
E_1	$[\{\wedge\}] \cup [\{1\}]$	$\{\wedge, 1\}$
E_2	$[\{\wedge\}]$	$\{\wedge\}$
N	$[\{\neg\}] \cup [\{0\}] \cup [\{1\}]$	$\{\neg, 1\}$
N_2	$[\{\neg\}]$	$\{\neg\}$
I	$[\{\mathrm{id}\}] \cup [\{0\}] \cup [\{1\}]$	$\{\mathrm{id}, 0, 1\}$
I_0	$[\{\mathrm{id}\}] \cup [\{0\}]$	$\{\mathrm{id}, 0\}$
I_1	$[\{\mathrm{id}\}] \cup [\{1\}]$	$\{\mathrm{id}, 1\}$
I_2	$[\{\mathrm{id}\}]$	$\{\mathrm{id}\}$

Figure 4: Table of all closed classes of Boolean functions, and their bases. Here, h_n denotes the function $h_n(x_1, \ldots, x_{n+1}) = \bigvee_{i=1}^{n+1} \bigwedge_{j=1, j\neq i}^{n+1} x_j$. See Definition 5.2 for the definition of $\mathsf{dual}(\cdot)$. The same table appears in [ABI+09, Table 1].

[BL88] Josh Cohen Benaloh and Jerry Leichter. Generalized secret sharing and monotone functions. In *Advances in Cryptology* (CRYPTO), pages 27–35, 1988.

[BST13] Eric Blais, Dominik Scheder, and Li-Yang Tan. Ajtai-gurevich redux. Manuscript. 2013.

[BT96] Nader H. Bshouty and Christino Tamon. On the Fourier spectrum of monotone functions. *J. ACM*, 43(4):747–770, 1996.

[Bul17] Andrei A. Bulatov. A dichotomy theorem for nonuniform csps. In Chris Umans, editor, *58th IEEE Annual Symposium on Foundations of Computer Science, FOCS 2017, Berkeley, CA, USA, October 15-17, 2017*, pages 319–330. IEEE Computer Society, 2017. 12

[Bul18] Andrei A. Bulatov. Constraint satisfaction problems: complexity and algorithms. *ACM SIGLOG News*, 5(4):4–24, 2018.

[CDM21] Arkadev Chattopadhyay, Rajit Datta, and Partha Mukhopadhyay. Lower bounds for monotone arithmetic circuits via communication complexity. In *STOC '21: 53rd Annual ACM SIGACT Symposium on Theory of Computing, Virtual Event, Italy, June 21-25, 2021*, pages 786–799. ACM, 2021.

[CHO+20] Lijie Chen, Shuichi Hirahara, Igor C. Oliveira, Ján Pich, Ninad Rajgopal, and Rahul Santhanam. Beyond Natural Proofs: Hardness Magnification and Locality. In *11th Innovations in Theoretical Computer Science Conference (ITCS 2020)*, volume 151, pages 70:1–70:48, 2020.

[CKR20] Bruno Pasqualotto Cavalar, Mrinal Kumar, and Benjamin Rossman. Monotone circuit lower bounds from robust sunflowers. In *LATIN 2020: Theoretical Informatics - 14th Latin American Symposium, São Paulo, Brazil, January 5-8, 2021, Proceedings*, volume 12118 of *Lecture Notes in Computer Science*, pages 311–322. Springer, 2020.

[CKS01] Nadia Creignou, Sanjeev Khanna, and Madhu Sudan. *Complexity classifications of Boolean constraint satisfaction problems*, volume 7 of *SIAM monographs on discrete mathematics and applications*. SIAM, 2001. 7, 9, 15

[COS17] Xi Chen, Igor C. Oliveira, and Rocco A. Servedio. Addition is exponentially harder than counting for shallow monotone circuits. In *STOC'17—Proceedings of the 49th Annual ACM SIGACT Symposium on Theory of Computing*, pages 1232–1245. ACM, New York, 2017.

[CZ16] Eshan Chattopadhyay and David Zuckerman. Explicit two-source extractors and resilient functions. In *Symposium on Theory of Computing* (STOC), pages 670–683, 2016.

[dRGR22] Susanna F. de Rezende, Mika Göös, and Robert Robere. Guest column: Proofs, circuits, and communication. *SIGACT News*, 53(1):59–82, 2022. 2

[FV98] Tomás Feder and Moshe Y. Vardi. The computational structure of monotone monadic SNP and constraint satisfaction: A study through datalog and group theory. *SIAM J. Comput.*, 28(1):57–104, 1998. 7, 13

[GGLR98] Oded Goldreich, Shafi Goldwasser, Eric Lehman, and Dana Ron. Testing monotonicity. In *Symposium on Foundations of Computer Science,* (FOCS), pages 426–435, 1998.

[GI12] Oded Goldreich and Rani Izsak. Monotone circuits: One-way functions versus pseudorandom generators. *Theory Comput.*, 8(1):231–238, 2012.

[GJW18] Mika Göös, Rahul Jain, and Thomas Watson. Extension complexity of independent set polytopes. *SIAM J. Comput.*, 47(1):241–269, 2018.

[GKRS19] Mika Göös, Pritish Kamath, Robert Robere, and Dmitry Sokolov. Adventures in monotone complexity and TFNP. In *10th Innovations in Theoretical Computer Science*, volume 124 of *LIPIcs. Leibniz Int. Proc. Inform.*, pages Art. No. 38, 19. Schloss Dagstuhl. Leibniz-Zent. Inform., Wadern, 2019. 2, 7, 11, 12

[GMOR15] Siyao Guo, Tal Malkin, Igor C. Oliveira, and Alon Rosen. The power of negations in cryptography. In *Theory of Cryptography Conference* (TCC), pages 36–65, 2015.

[GS92] Michelangelo Grigni and Michael Sipser. Monotone complexity. In *Proceedings of the London Mathematical Society Symposium on Boolean Function Complexity*, page 57–75, USA, 1992. Cambridge University Press.

[Hås86] Johan Håstad. Almost optimal lower bounds for small depth circuits. In *Proceedings of the 18th Annual ACM Symposium on Theory of Computing, May 28-30, 1986, Berkeley, California, USA*, pages 6–20, 1986.

[Hir22] Shuichi Hirahara. NP-hardness of learning programs and partial MCSP. In *Symposium on Foundations of Computer Science* (FOCS), 2022.

[HWWY94] Johan Håstad, Ingo Wegener, Norbert Wurm, and Sang-Zin Yi. Optimal depth, very small size circuits for symmetric functions in AC^0. *Inf. Comput.*, 108(2):200–211, 1994.

[IKL+19] Christian Ikenmeyer, Balagopal Komarath, Christoph Lenzen, Vladimir Lysikov, Andrey Mokhov, and Karteek Sreenivasaiah. On the complexity of hazard-free circuits. *J. ACM*, 66(4):25:1–25:20, 2019.

[JCG97] Peter Jeavons, David Cohen, and Marc Gyssens. Closure properties of constraints. *J. ACM*, 44(4):527–548, jul 1997. 4

[Jea98] Peter Jeavons. On the algebraic structure of combinatorial problems. *Theor. Comput. Sci.*, 200(1-2):185–204, 1998. 3, 5, 14

[JLL76] Neil D. Jones, Y. Edmund Lien, and William T. Laaser. New problems complete for nondeterministic log space. *Math. Syst. Theory*, 10:1–17, 1976. 15

[Joh03] Jan Johannsen. The complexity of satisfiability problems with two occurrences. Unpublished Manuscript, 2003.

[Juk12] Stasys Jukna. *Boolean Function Complexity - Advances and Frontiers*. Springer, 2012. 14

[Kra97] Jan Krajíček. Interpolation theorems, lower bounds for proof systems, and independence results for bounded arithmetic. *J. Symbolic Logic*, 62(2):457–486, 1997.

[KS92] Bala Kalyanasundaram and Georg Schnitger. The probabilistic communication complexity of set intersection. *SIAM J. Discret. Math.*, 5(4):545–557, 1992.

[Kup21] Denis Kuperberg. Positive first-order logic on words. In *Symposium on Logic in Computer Science* (LICS), pages 1–13, 2021.

[Kup22] Denis Kuperberg. Positive first-order logic on words and graphs. *CoRR*, abs/2201.11619, 2022.

[KW93] Mauricio Karchmer and Avi Wigderson. On span programs. In *Structure in Complexity Theory Conference* (CCC), pages 102–111, 1993.

[LT09] Benoît Larose and Pascal Tesson. Universal algebra and hardness results for constraint satisfaction problems. *Theor. Comput. Sci.*, 410(18):1629–1647, 2009.

[Oko82] E. A. Okol'nishnikova. The effect of negations on the complexity of realization of monotone Boolean functions by formulas of bounded depth. *Metody Diskret. Analiz.*, (38):74–80, 1982.

[Oli15] Igor C. Oliveira. Unconditional lower bounds in complexity theory, 2015. (PhD Thesis, Columbia University).

[OSS19] Igor C. Oliveira, Rahul Santhanam, and Srikanth Srinivasan. Parity helps to compute majority. In *34th Computational Complexity Conference, CCC 2019, July 18-20, 2019, New Brunswick, NJ, USA*, pages 23:1–23:17, 2019.

[Pos41] Emil L. Post. *The Two-Valued Iterative Systems of Mathematical Logic. (AM-5)*. Princeton University Press, 1941.

[PR17] Toniann Pitassi and Robert Robere. Strongly exponential lower bounds for monotone computation. In *Symposium on Theory of Computing* (STOC), pages 1246–1255, 2017.

[Pud97] Pavel Pudlák. Lower bounds for resolution and cutting plane proofs and monotone computations. *J. Symb. Log.*, 62(3):981–998, 1997.

[Qui53] W. V. Quine. Two theorems about truth functions. *Bol. Soc. Mat. Mexicana*, 10:64–70, 1953.

[Raz85a] A. A. Razborov. Lower bounds on the monotone complexity of some Boolean functions. *Dokl. Akad. Nauk SSSR*, 281(4):798–801, 1985.

[Raz85b] Alexander A Razborov. Lower bounds on monotone complexity of the logical permanent. *Mathematical Notes of the Academy of Sciences of the USSR*, 37(6):485–493, 1985. 2, 12

[Rei05] Omer Reingold. Undirected st-connectivity in log-space. In Harold N. Gabow and Ronald Fagin, editors, *Proceedings of the 37th Annual ACM Symposium on Theory of Computing, Baltimore, MD, USA, May 22-24, 2005*, pages 376–385. ACM, 2005.

[Rez23] Susanna Rezende. Personal communication, 2023. 12

[RM99] Ran Raz and Pierre McKenzie. Separation of the monotone NC hierarchy. *Combinatorica*, 19(3):403–435, 1999. 7, 12, 14

[Ros08a] Benjamin Rossman. Homomorphism preservation theorems. *J. ACM*, 55(3):Art. 15, 53, 2008.

[Ros08b] Benjamin Rossman. On the constant-depth complexity of k-clique. In *Symposium on Theory of Computing* (STOC), pages 721–730, 2008.

[Ros17a] Benjamin Rossman. An entropy proof of the switching lemma and tight bounds on the decision-tree size of AC^0. 2017.

[Ros17b] Benjamin Rossman. An improved homomorphism preservation theorem from lower bounds in circuit complexity. In *Innovations in Theoretical Computer Science Conference* (ITCS), pages 27:1–27:17, 2017.

[RPRC16] Robert Robere, Toniann Pitassi, Benjamin Rossman, and Stephen A. Cook. Exponential lower bounds for monotone span programs. In *IEEE 57th Annual Symposium on Foundations of Computer Science, FOCS 2016, 9-11 October 2016, Hyatt Regency, New Brunswick, New Jersey, USA*, pages 406–415. IEEE Computer Society, 2016.

[RW92] Ran Raz and Avi Wigderson. Monotone circuits for matching require linear depth. *J. ACM*, 39(3):736–744, 1992.

[Sch78] Thomas J. Schaefer. The complexity of satisfiability problems. In *Proceedings of the 10th Annual ACM Symposium on Theory of Computing, May 1-3, 1978, San Diego, California, USA*, pages 216–226. ACM, 1978. 3, 5, 12, 14, 15

[Sto95] Alexei P. Stolboushkin. Finitely monotone properties. In *Symposium on Logic in Computer Science* (LICS), pages 324–330, 1995.

[Tar88] É. Tardos. The gap between monotone and nonmonotone circuit complexity is exponential. *Combinatorica*, 8(1):141–142, 1988. 2, 12

[Urq87] Alasdair Urquhart. Hard examples for resolution. *J. ACM*, 34(1):209–219, 1987.

[Val80] Leslie G. Valiant. Negation can be exponentially powerful. *Theor. Comput. Sci.*, 12:303–314, 1980.

[Zhu17] Dmitriy Zhuk. A proof of CSP dichotomy conjecture. In Chris Umans, editor, *58th IEEE Annual Symposium on Foundations of Computer Science, FOCS 2017, Berkeley, CA, USA, October 15-17, 2017*, pages 331–342. IEEE Computer Society, 2017. 12